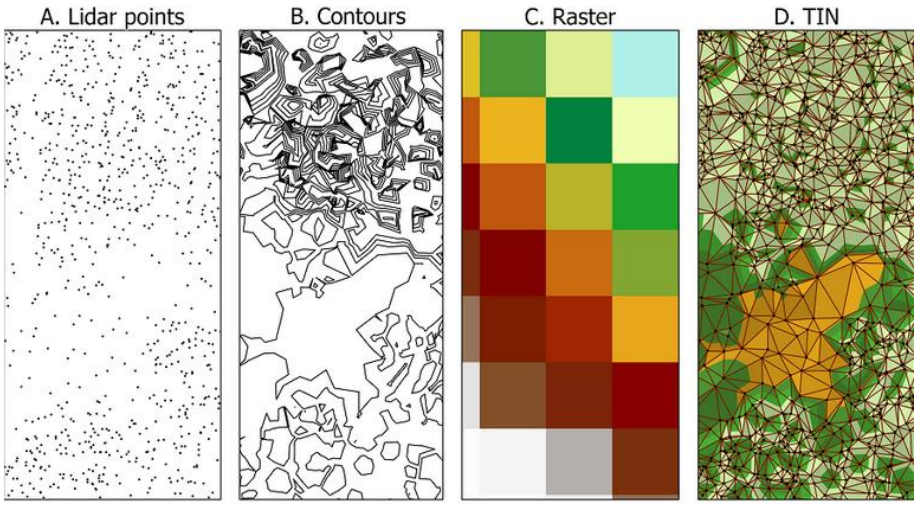
Terrain Analysis

Lecture content

- What is terrain analysis?
- Data sources, types and formats.
- Slope, aspect, curvature and topographic profiles.
- Catchment and hydrology.
- Line of sight and viewsheds.
- Solar radiation mapping.

What is Terrain Analysis

- Complexity: spheroid < geoid < land surface (terrain).
- Using topographic data to derive terrain features.
  - the morphology of terrain surface.
  - the influence of terrain on environmental processes.
- Contours, spot heights, DEM (raster) and TIN (vector).



different ways of representing the same continuous surface at the same scale and extent

https://gistbok.ucgis.org/bok-topics/triangular-irregular-network-tin-models00

Main digital terrain data sources

- Ground surveys.
- Photogrammetry.
- Radar altimetry.
- Interferometric synthetic aperture radar (InSAR).
- Airborne laser scanning (LiDAR).
- Most of these are in the form of, or can be readily transformed to DEM.

Applications of terrain analysis

Terrain analysis is about how the shape of the land influences or reflects natural processes, hazards, ecosystems, and human influences.

Mapping drainage networks — identifying channels, flow paths, and watershed boundaries for river management.

Flood modelling — locating floodplains, flow bottlenecks, and overtopping points.

Erosion risk assessment —to predict where soil loss is likely.

Wetland and groundwater recharge mapping — identifying depressions, low-slope zones, and convergent flow areas.

Geomorphological mapping of ridges, valleys, terraces, and scarps, dunes.

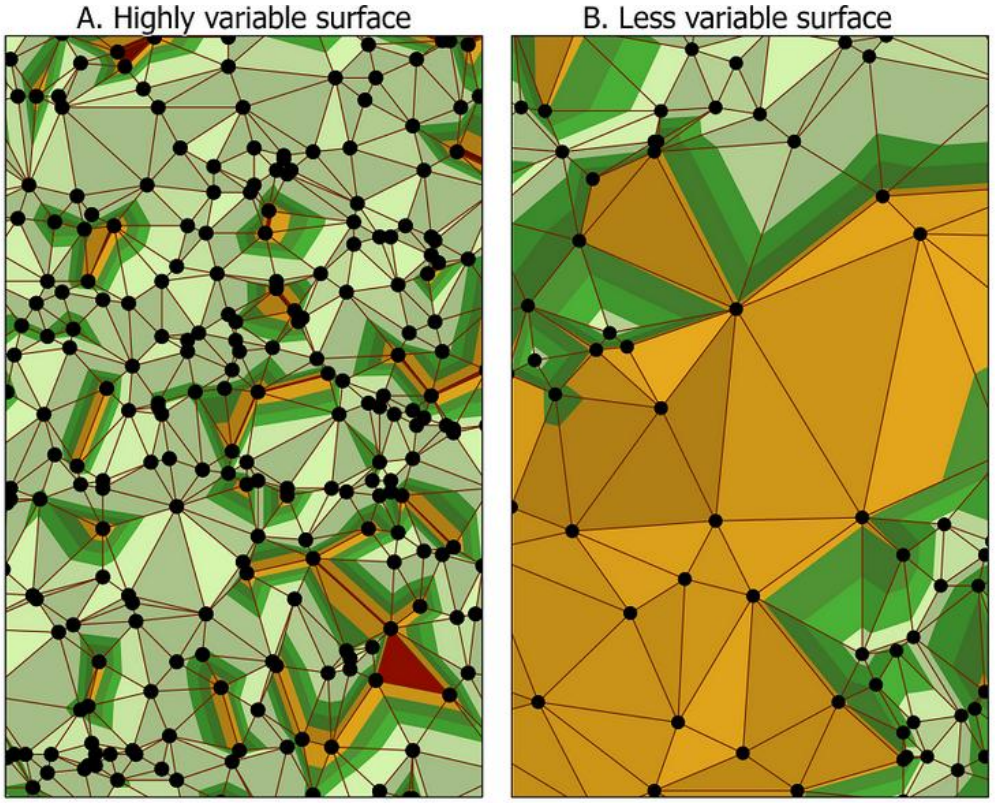
Shape parameters of volcanic landforms (e.g. subglacial edifies on Mars)

Detecting landslide-prone slopes.

Mapping glacial features such as drumlins, moraines, and meltwater channels.

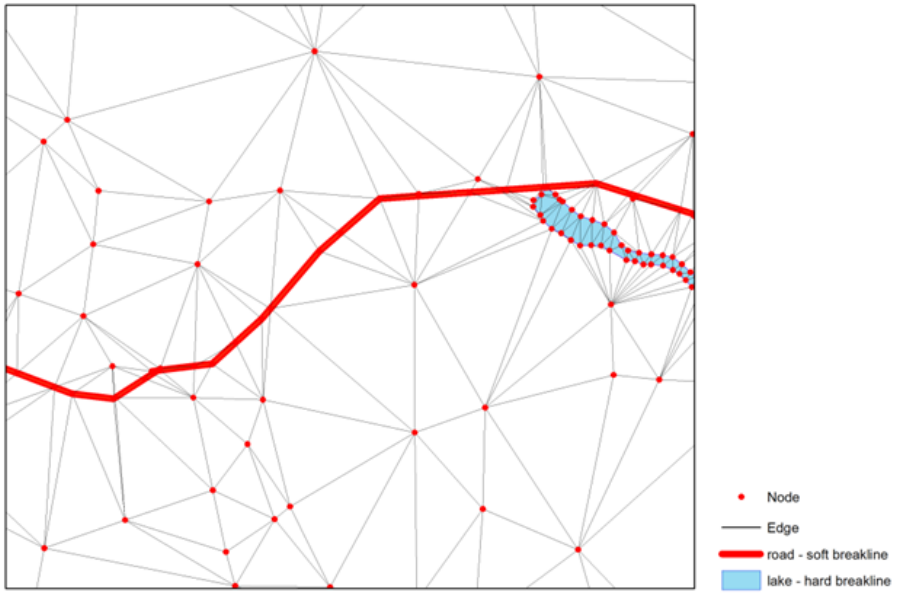
Triangulated irregular networks

- TIN - using contiguous, non-overlapping triangles linking 3 adjacent elevation points.
- Independent of data scale = flexible in level of detail.
- Efficient in terms of data storage.
- TIN less useful than DEM when surface is flat (e.g. plateau, deserts).



TIN with hard and soft breaklines

A breakline is a line feature with elevation values that forces the TIN surface to follow a specific shape. They prevent the interpolation from smoothing over important terrain edges.



- *Hard breakline* – a discontinuity in slope of a surface.
- *Soft breakline* – does not change local slope of terrain surface.
  - Hard breaklines keep water inside channels; soft breaklines guides the flow along the deepest parts of channels.
  - Hard breaklines preserve scarps; soft breaklines maintain natural curvature.

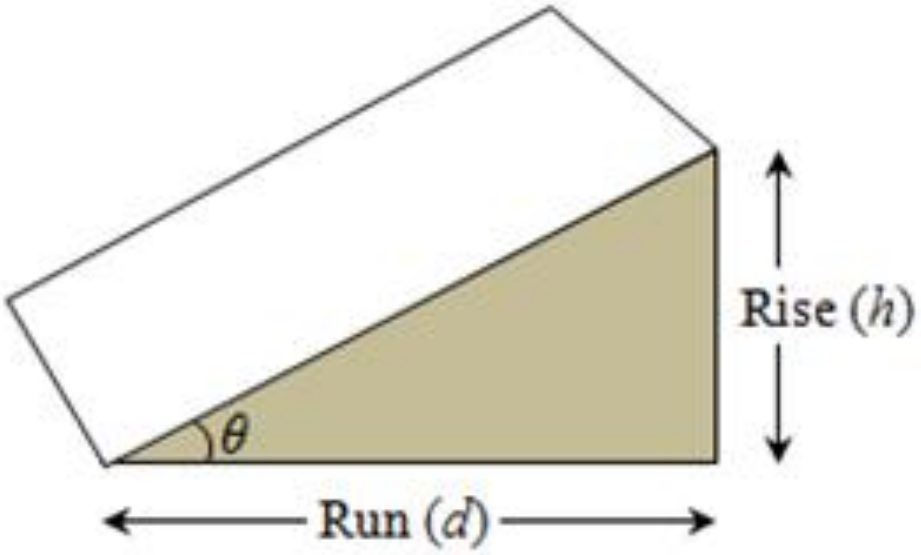
Primary terrain attributes

- Elevation, slope, aspect, plan and profile curvature are primary terrain attributes.
- Flow direction, flow accumulation, flow length, flow order, drainage network and catchment area are derived (hydrological) characteristics of a terrain surface.
- In GIS, they are mainly derived from DEM or TIN and represented in raster layers.

Secondary terrain attributes

- Secondary attributes are computed from two or more primary attributes.
- Quantify the role played by topography in redistributing water in the landscape and in modifying the amount of solar radiation received at the surface.
- These attributes affect soil characteristics, distribution and abundance of soil water, susceptibility of landscapes to erosion by water, and the distribution and abundance of flora and fauna.

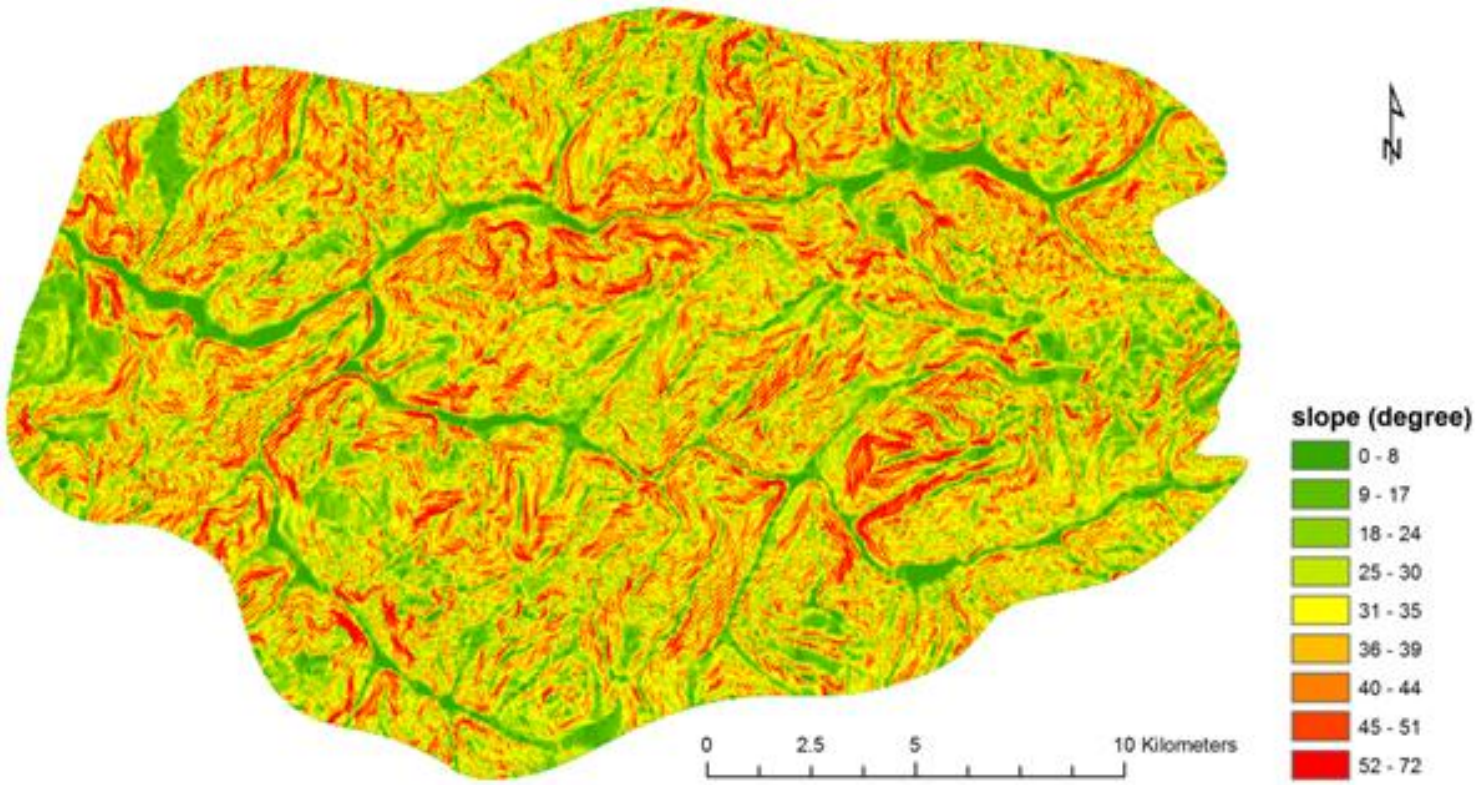
Slope



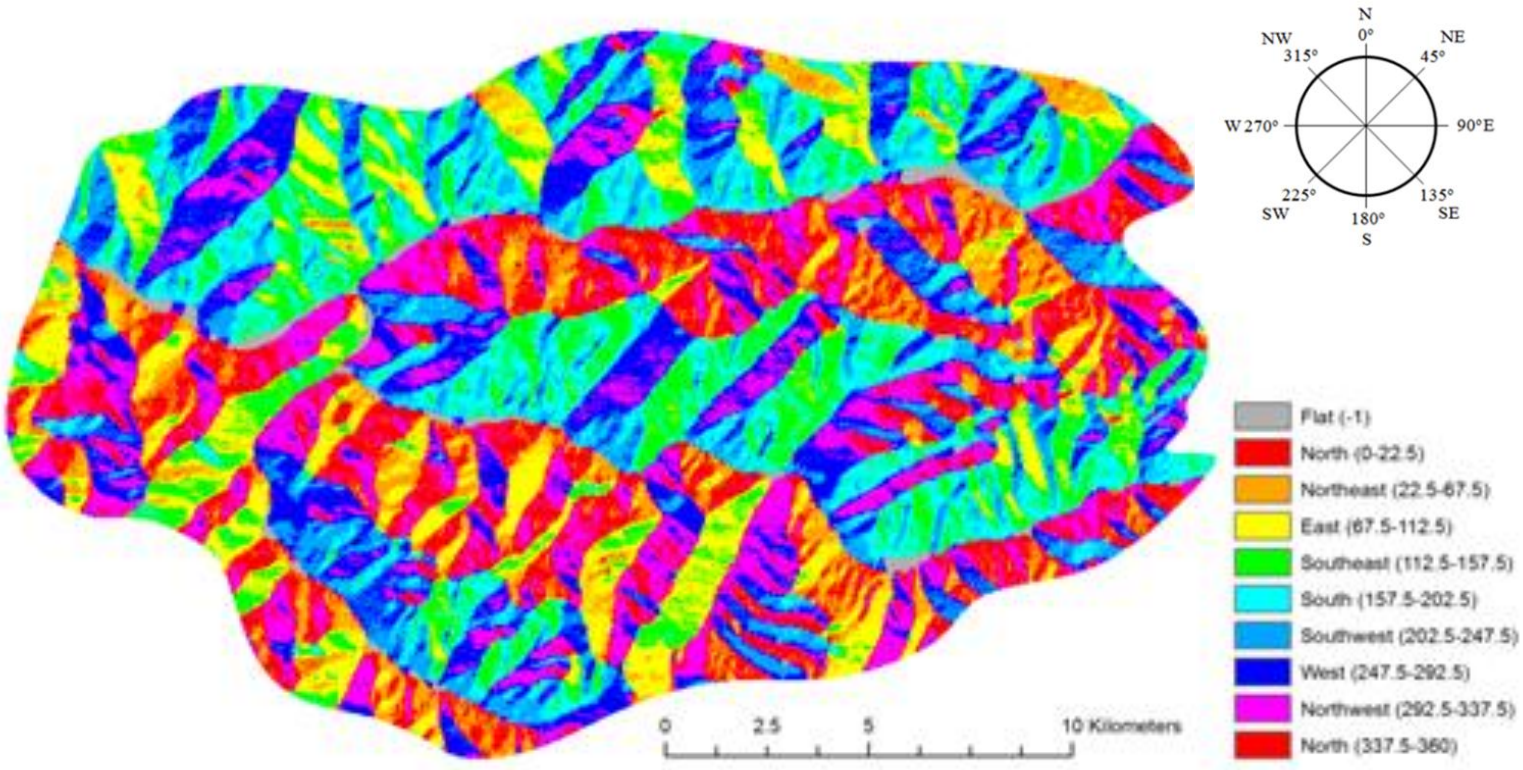
Percent slope: $\theta = 100 \times (h / d)$ (%)

Degree slope: $\theta = \arctan(h / d)$ (degree)

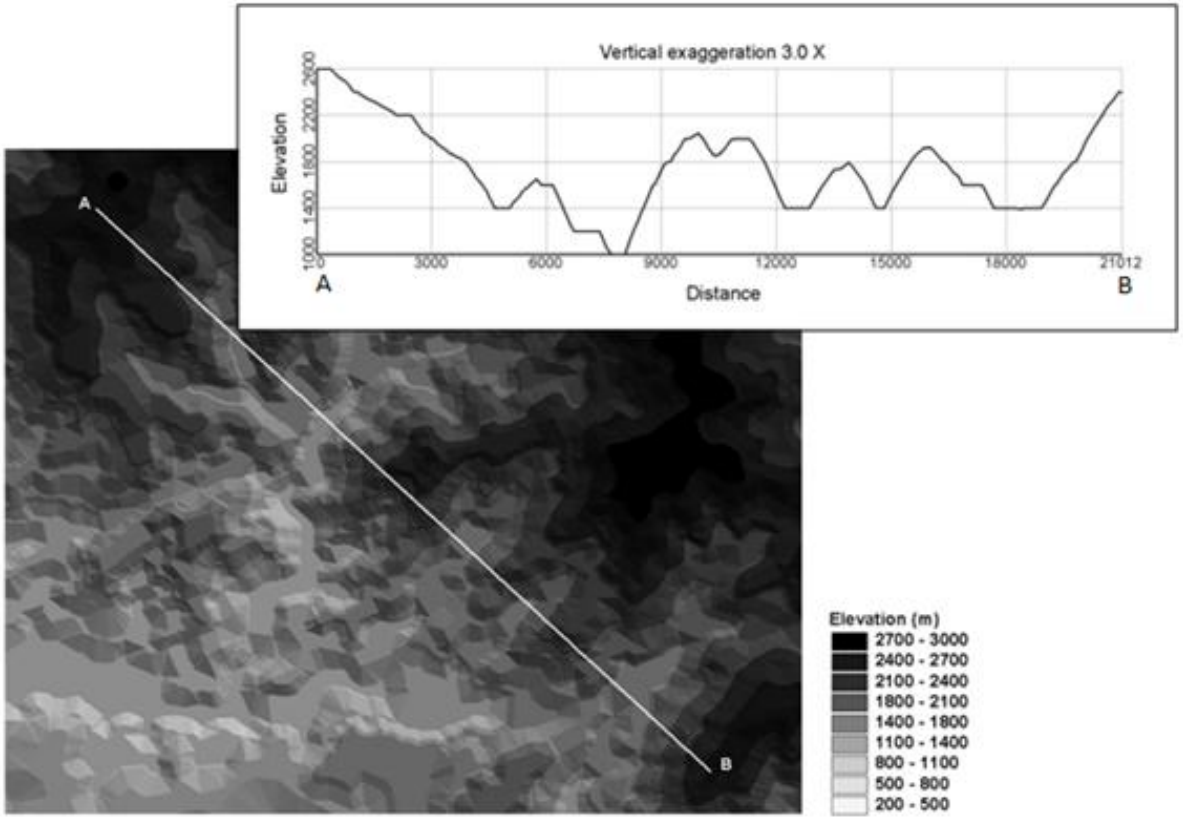
Slope map


N
slope (degree)
0 - 8
9 - 17
18 - 24
25 - 30
31 - 35
36 - 39
40 - 44
45 - 51
52 - 72
0
2.5
5
10 Kilometers

Aspect map

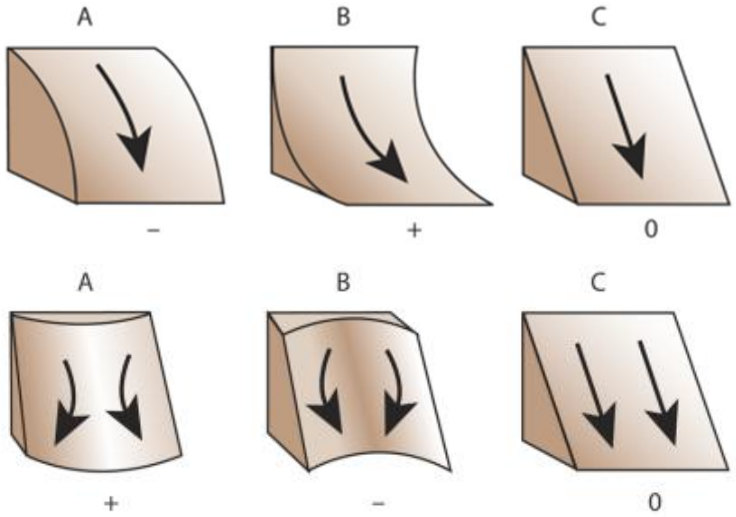

N
0°
NE
45°
90°E
135°
SE
180°
S
225°
SW
W 270°
NW
315°
0
2.5
5
10 Kilometers
Flat (-1)
North (0-22.5)
Northeast (22.5-67.5)
East (67.5-112.5)
Southeast (112.5-157.5)
South (157.5-202.5)
Southwest (202.5-247.5)
West (247.5-292.5)
Northwest (292.5-337.5)
North (337.5-360)

Terrain profile



Surface curvature

- Amount by which a surface deviates from being flat.
- **Profile curvature** is the rate of change of slope in the direction of the maximum slope.
- **Plan curvature** is the curvature perpendicular to the direction of the maximum slope

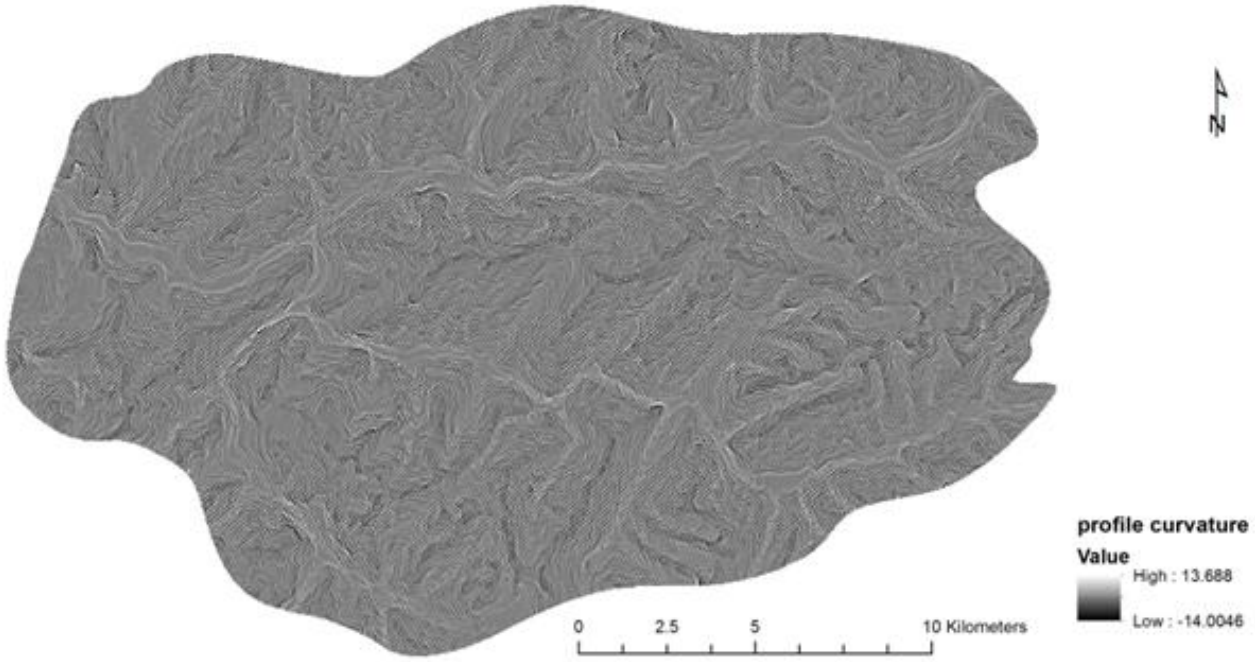


Profile curvature = maximum slope of this surface

Planform curvature = convergence/divergence

https://www.esri.com/arcgis-blog/products/product/imagery/understanding-curvature-rasters/

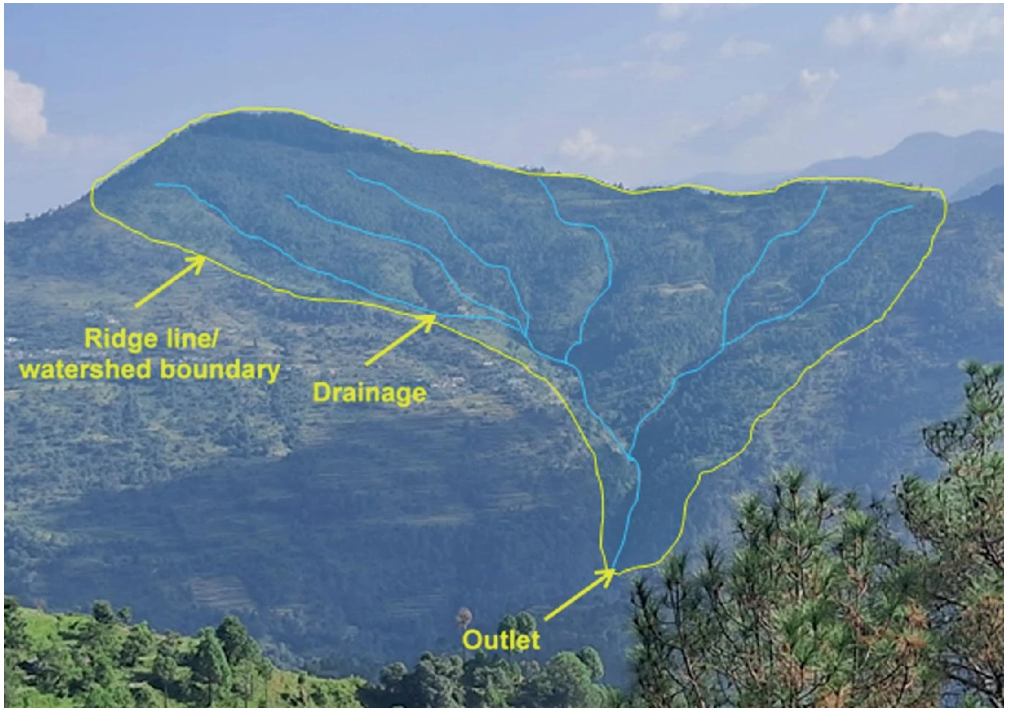
Profile curvature map



Slope influences the overall rate of movement downslope, **aspect** determines the direction of flow, **profile curvature** affects the flow acceleration and erosion/deposition rate, and **plan curvature** affects the convergence and divergence of flow.

Watershed analysis

- Models the movement of water over the land surface.
- Track pollution sources to locate potentially affected streams e.g. agrochemicals, outlet pipes from factories.
- Estimate stream flow for ungauged areas.
- Predict water quality.
- Analyse historic and future climate predictions.
- Understand sedimentation processes.
- Characterise landforms.



Hydrologically correct DEM

- Checking the DEM for sinks (hydrological conditioning).
- Areas which are pits, surrounded by higher elevation values.
- Whilst these can occur naturally, often they result from errors in the input DEM.
- Bridges, roads, quarries, sinkholes or misrepresentation of narrow-sided channels can all result in sinks.

450	446	441	451	454
447	441	440	446	451
440	339	431	440	445
438	435	422	431	439
431	429	414	422	424

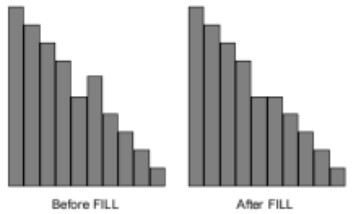


Figure 3.4a: Chopping off tall cells

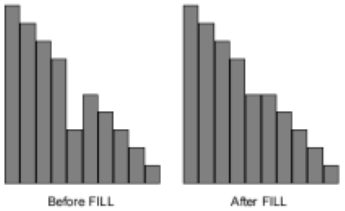
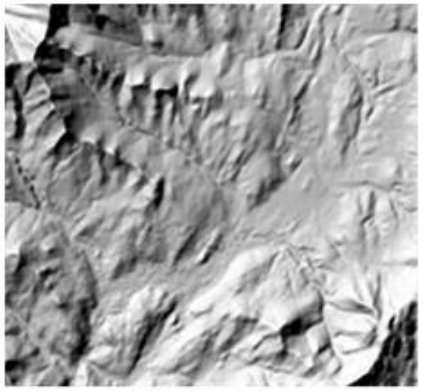
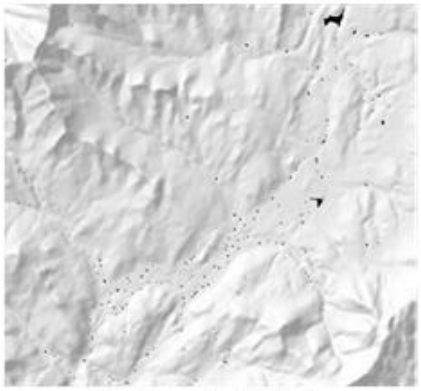


Figure 3.4b: Filing in sinks

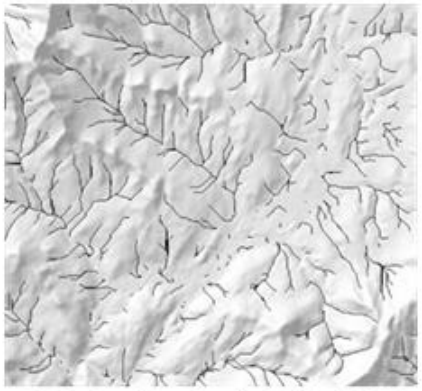
Hydrologically correct DEM



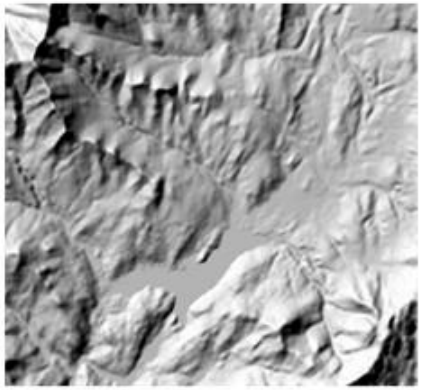
(a) DEM with sinks



(b) Sinks marked in black



(c) Drainage lines from the DEM with sinks

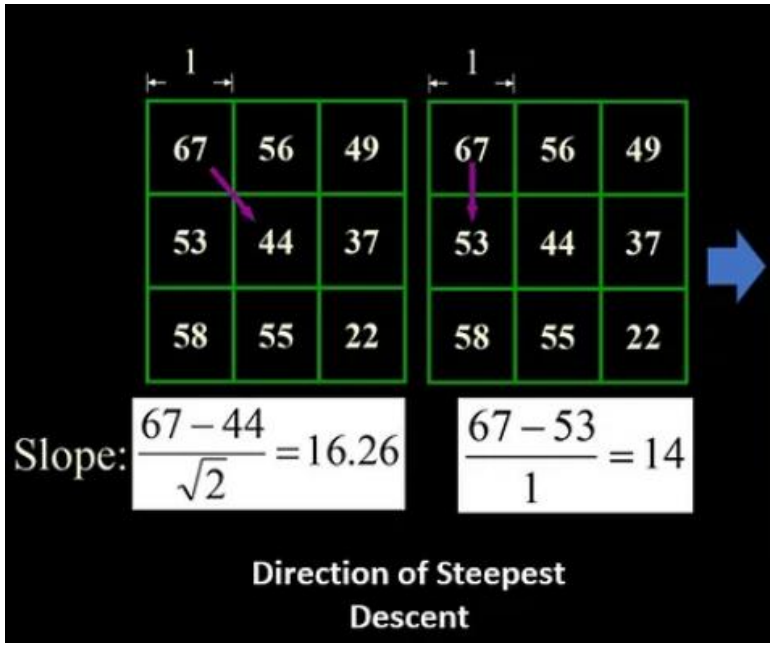


(d) DEM without sinks



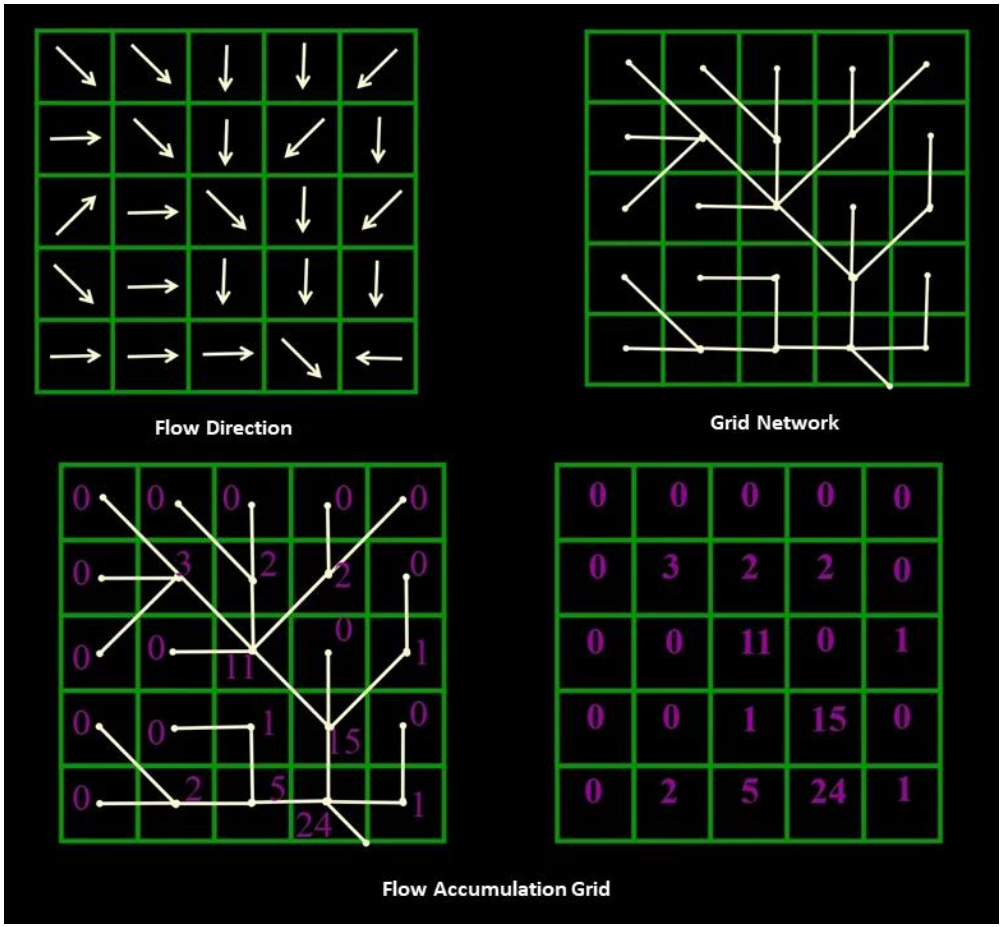
(e) Drainage lines from the DEM without sinks

Flow direction

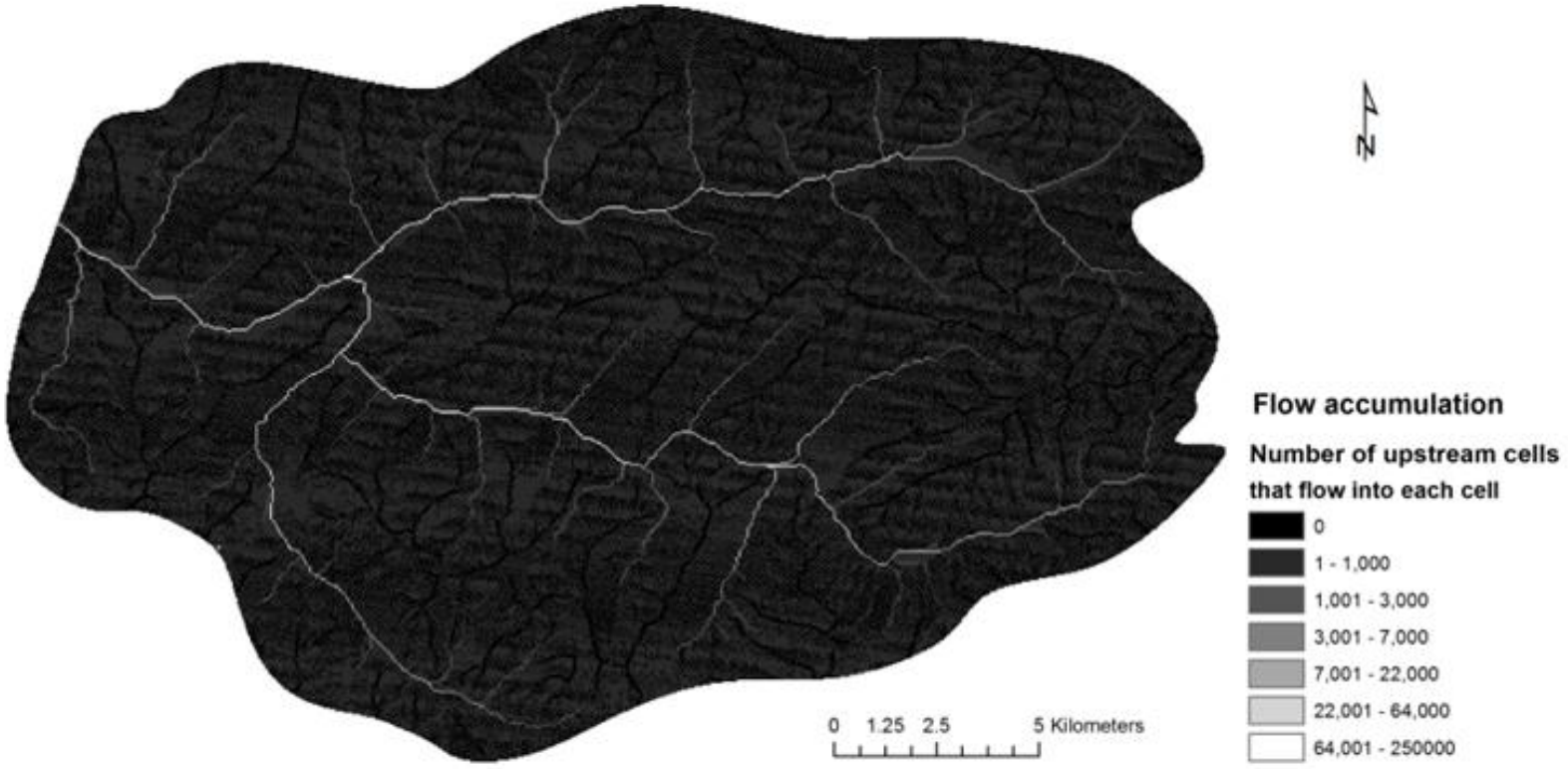


DEM – starting raster

flow accumulation – derived raster



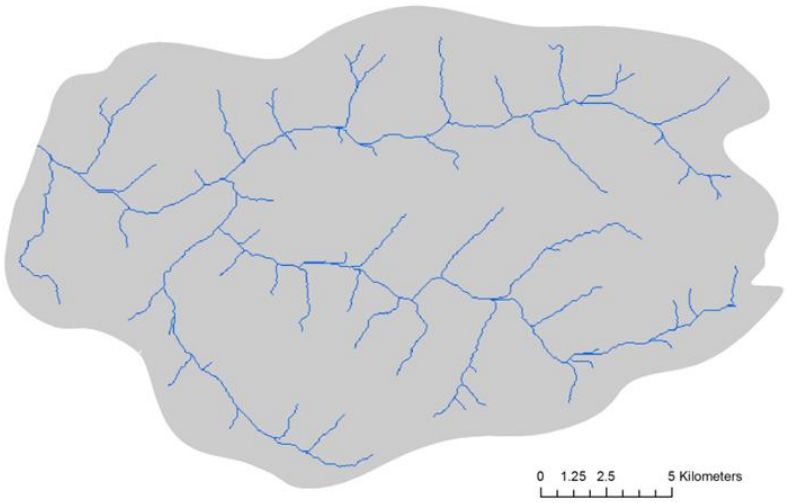
Flow accumulation map



High values = stream channels; zero values = ridgelines

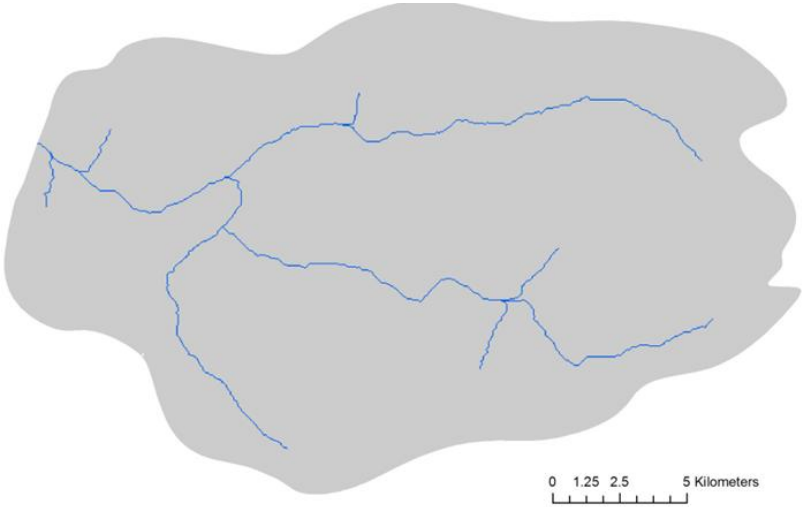
Drainage network

Threshold values : 1000 cells



(a)

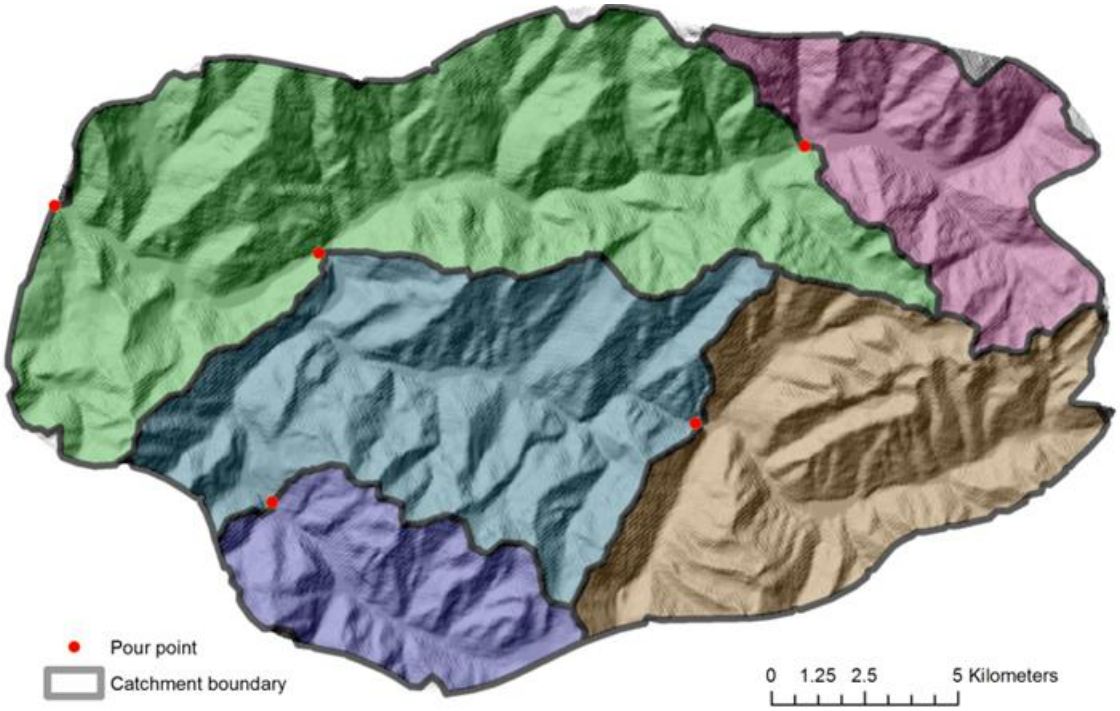
Threshold values : 7000 cells



(b)

- Threshold values are numerical cut-offs used to decide where surface runoff becomes a stream. Thresholds convert continuous flow accumulation into a discrete drainage network.

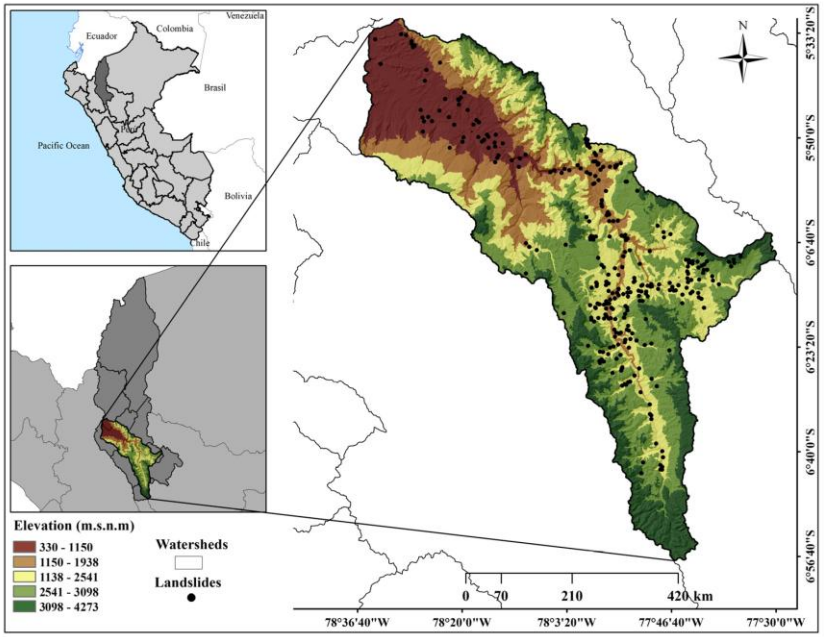
Catchment boundary



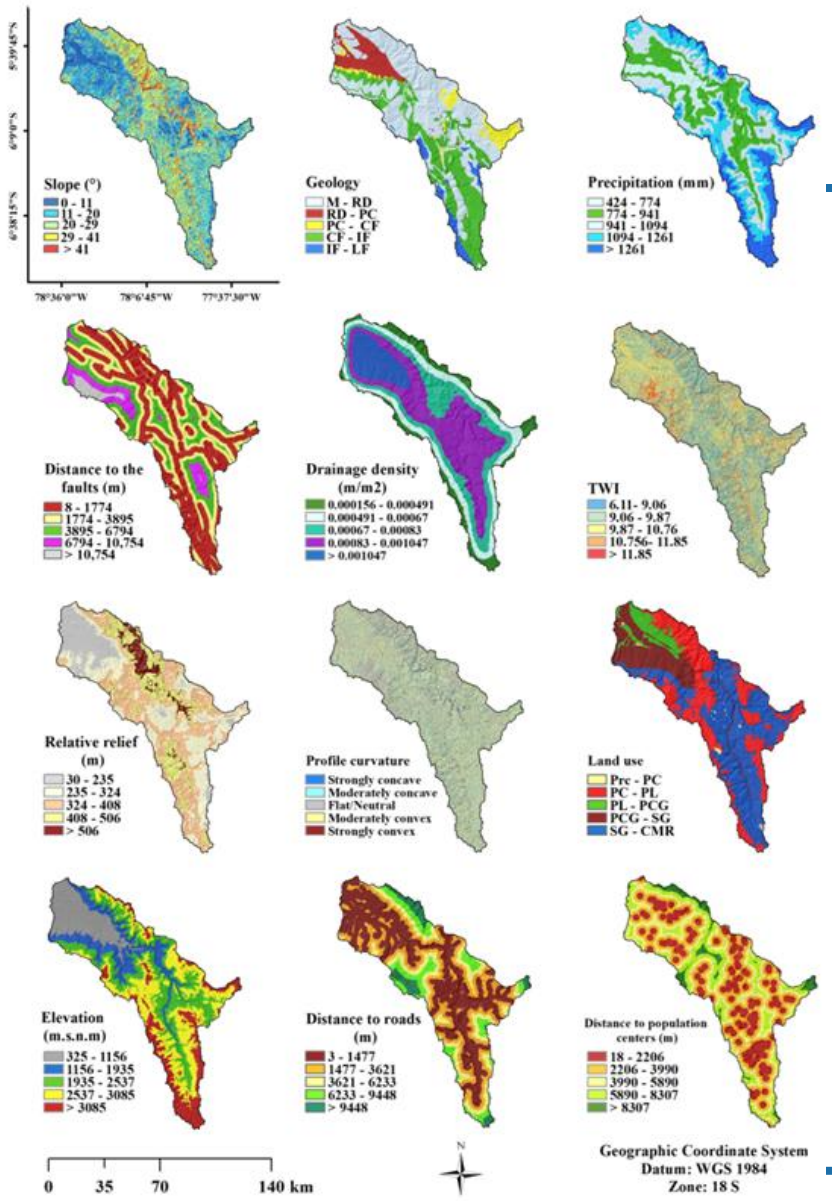
A pour point in GIS is the specific point where water flows out of a catchment. It is the lowest point on a watershed boundary and the location where all upstream flow converges and “pours” into the next downstream area.

Landslide risk criteria

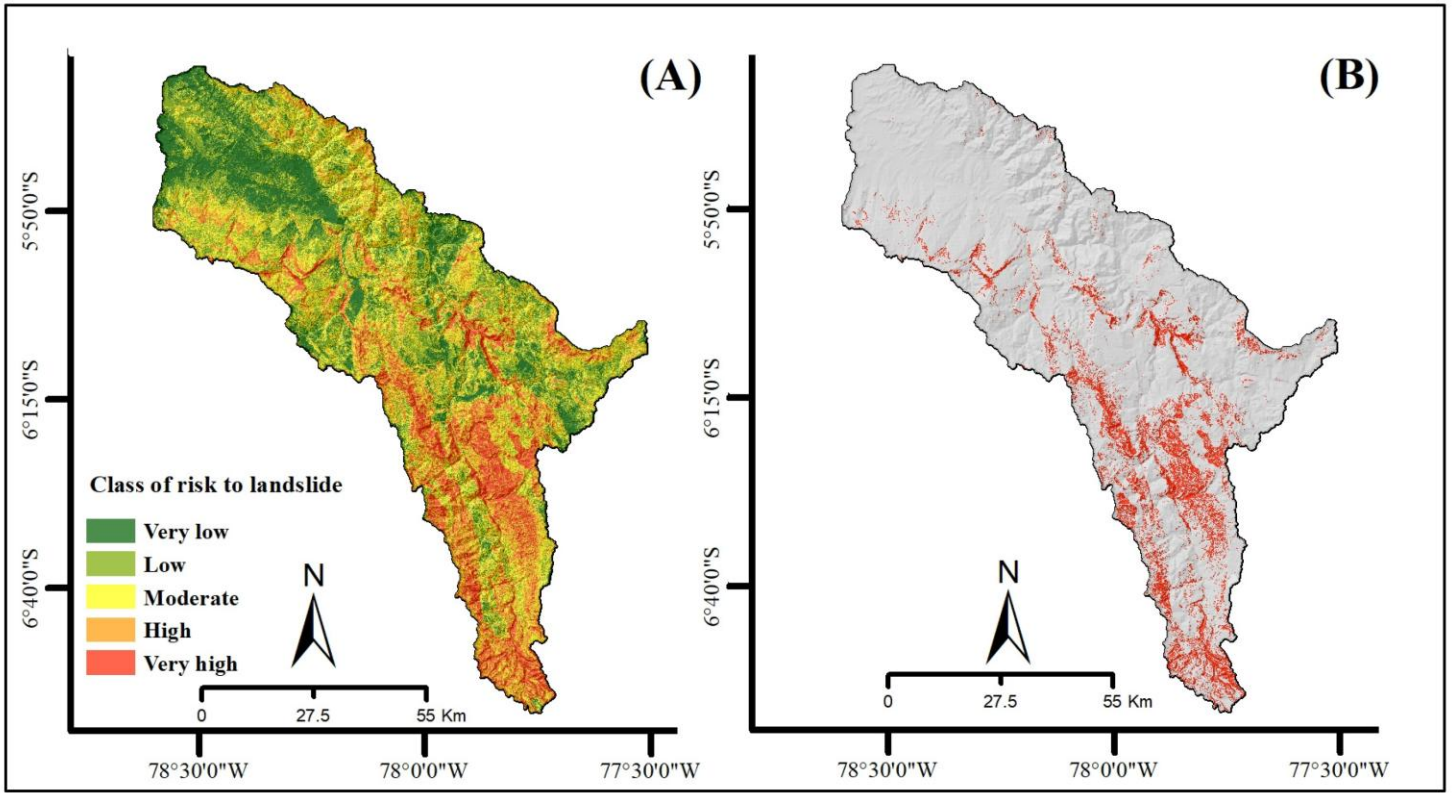
Utcubamba River Basin, Peru



Rivera et al. (2025)
https://doi.org/10.3390/app15179423

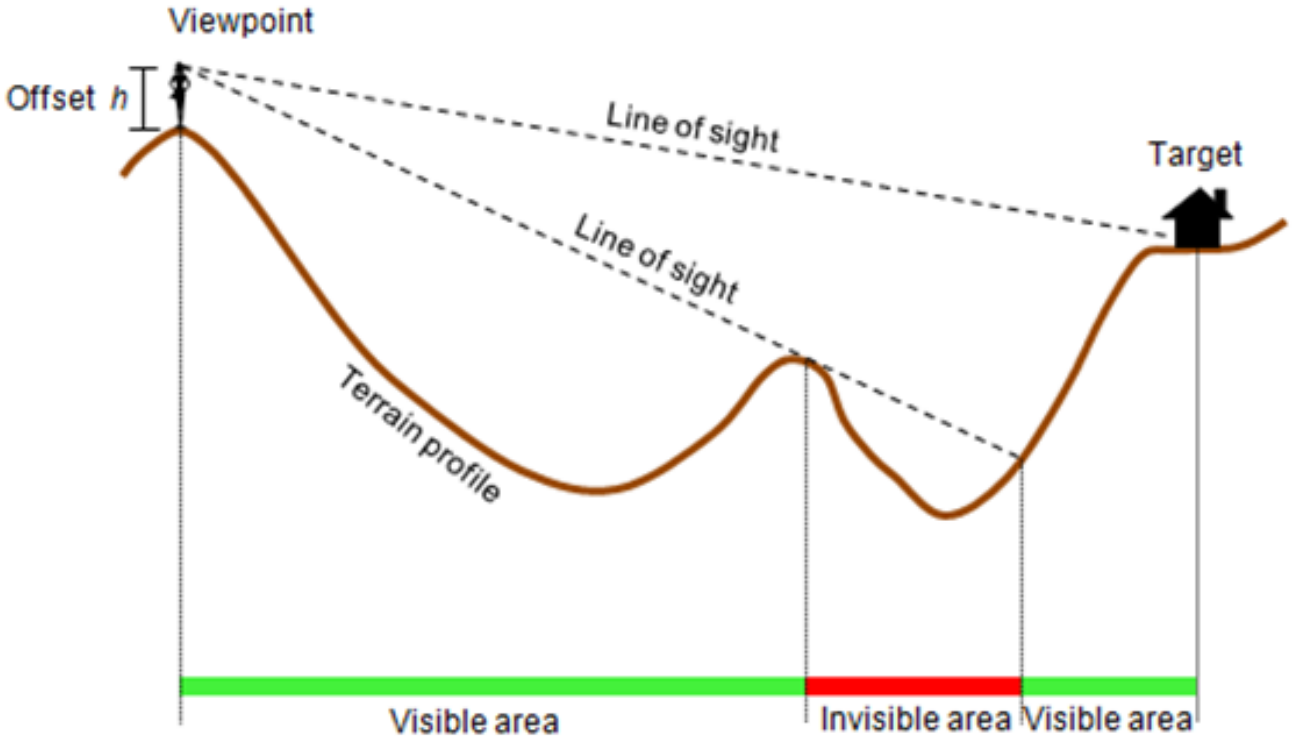


Landslide risk zoning and mapping



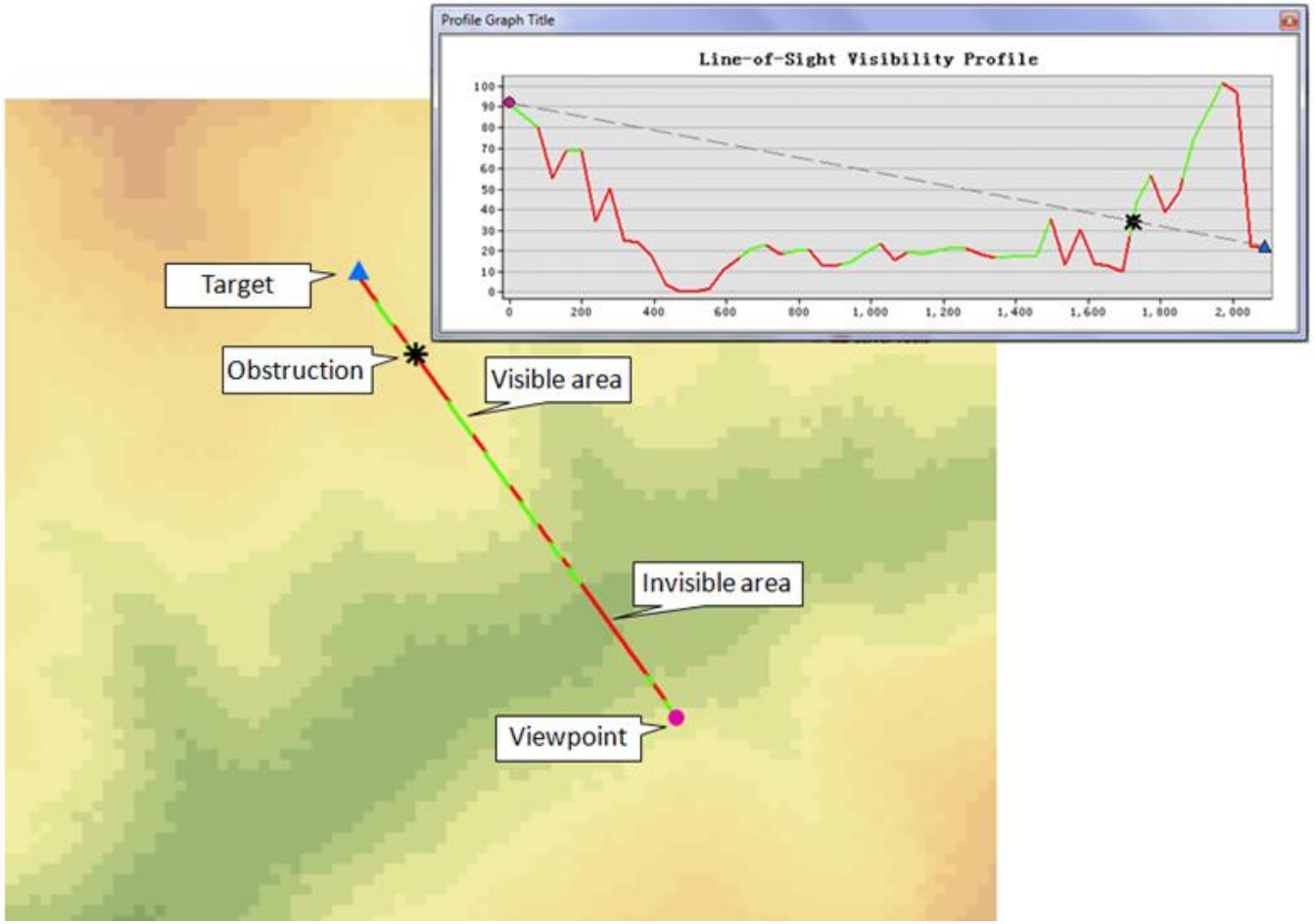
Rivera et al. (2025)
https://doi.org/10.3390/app15179423

Visibility analysis: Line-of-sight

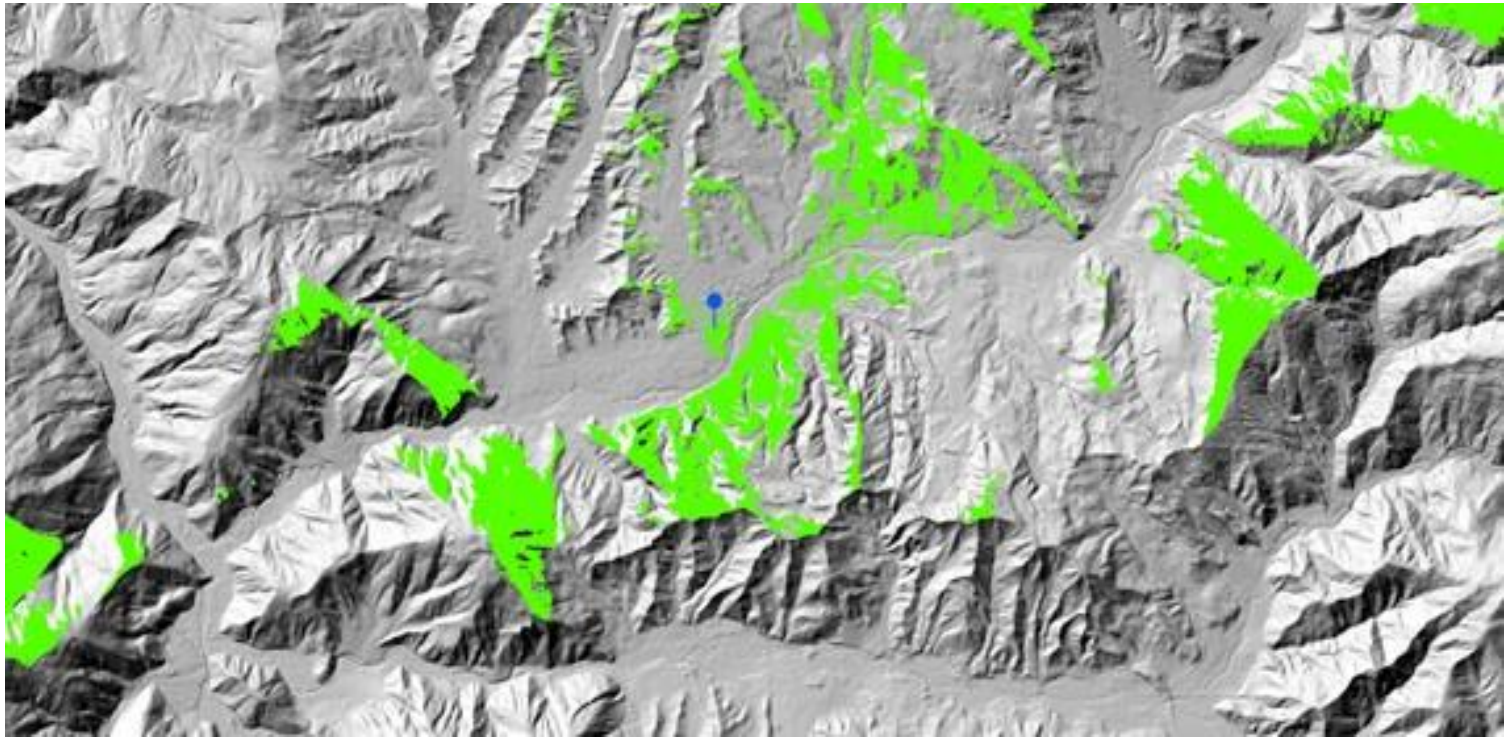


Line of sight - Draws a line between the observer point and the target point to indicate the visibility (visible or obstructed) along the line between the given points.

Typical output of line-of-sight analysis



Visibility analysis: Viewshed

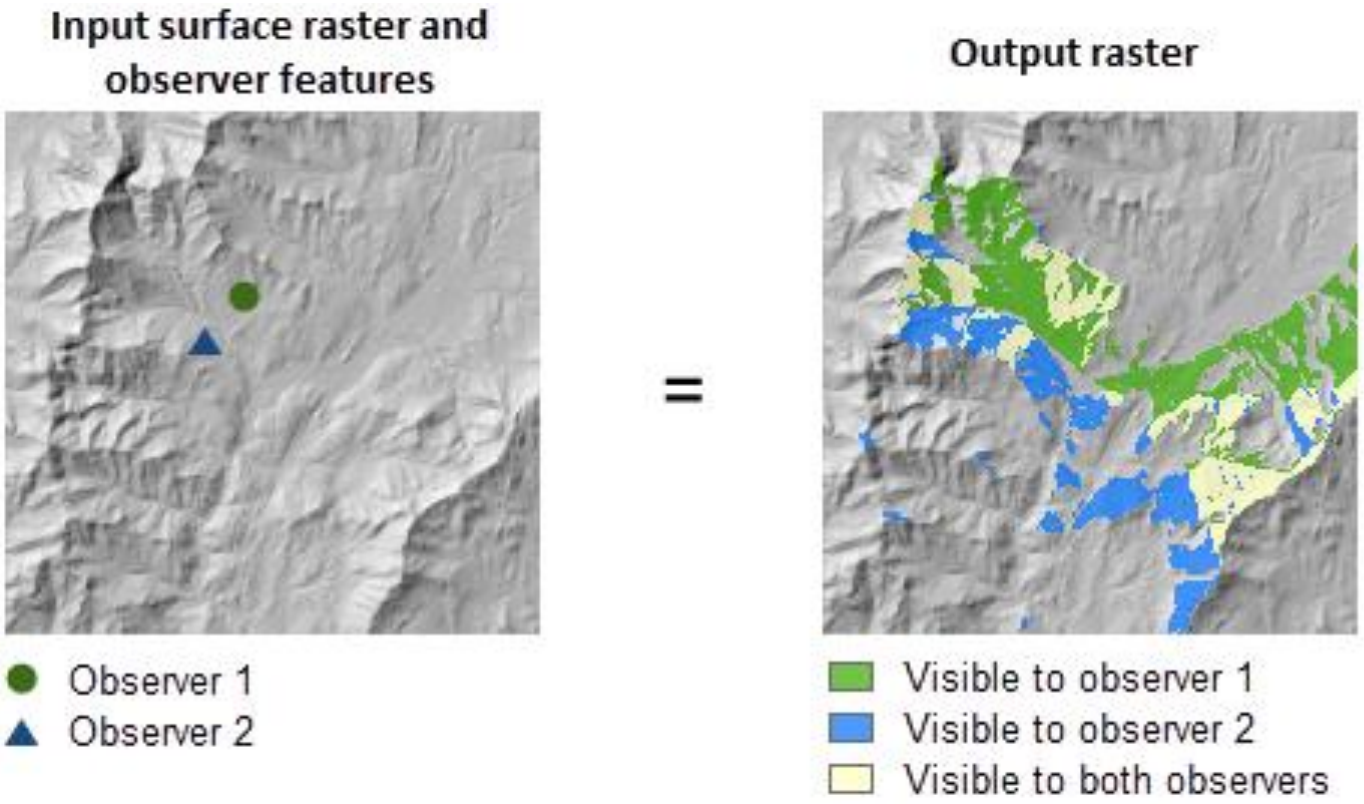


Green = visible from blue point.

Viewshed - Shows the visibility (visible or obstructed) of an area from a specified viewpoint.

https://giscrack.com/viewshed-analysis-using-the-geodesic-viewshed-tool-in-arcgis-pro/

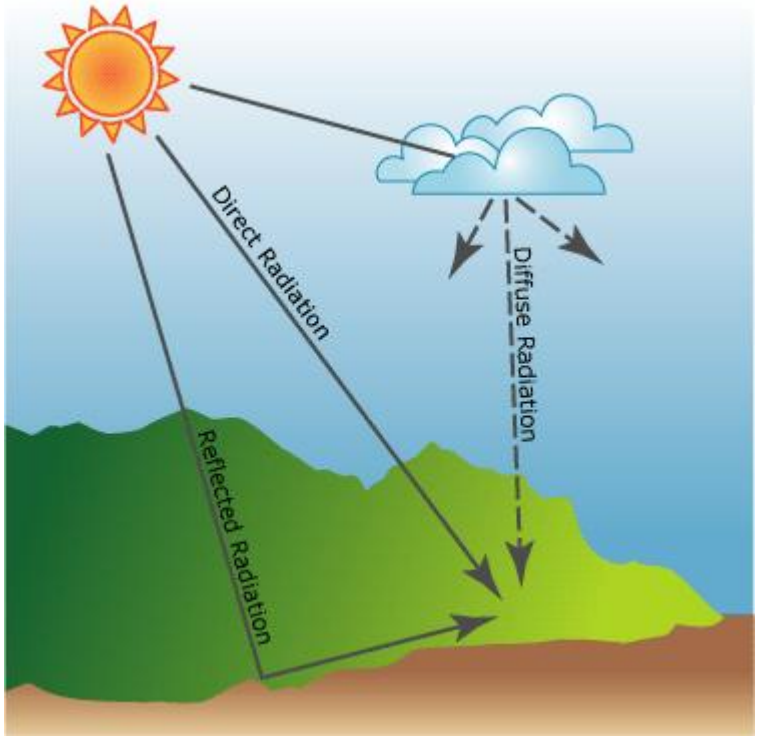
Visibility analysis: Viewshed 2



https://pro.arcgis.com/en/pro-app/3.4/tool-reference/spatial-analyst/viewshed-2.htm

Solar radiation mapping

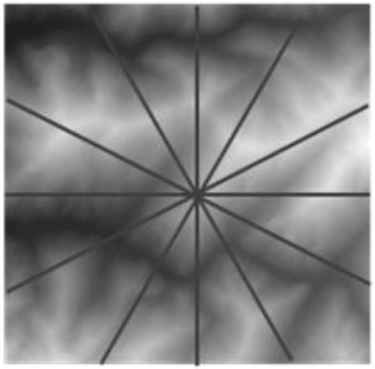
- Estimate spatial and temporal distribution of solar radiation on the Earth's surface.
- Solar radiation is radiant energy emitted by the Sun.
- **Direct radiation** propagates unimpeded in a direct line from the Sun to the receiving surface.
- **Diffuse radiation** is scattered by clouds and dust.
- ***Reflected radiation*** is the direct and diffuse radiation reflected by nearby terrain towards the location of interest.
- Total or **global** solar radiation is the sum of the direct, diffuse and reflected radiation.



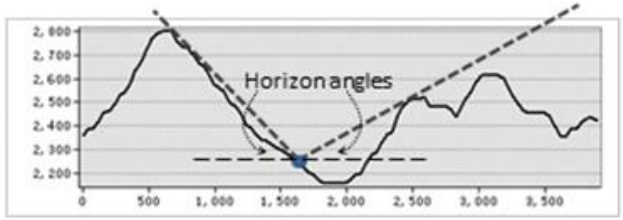
https://desktop.arcgis.com/en/arcmap/latest/tools/spatial-analyst-toolbox/modeling-solar-radiation.htm

Hemispherical viewshed

- Upward–looking hemispherical viewshed model (DTM).
- Maximum angle of sky obstruction in all directions.
- Used in combination with sunmaps and skymaps.
- Calculate solar radiation for each location.



DEM and directions for calculating horizon angles

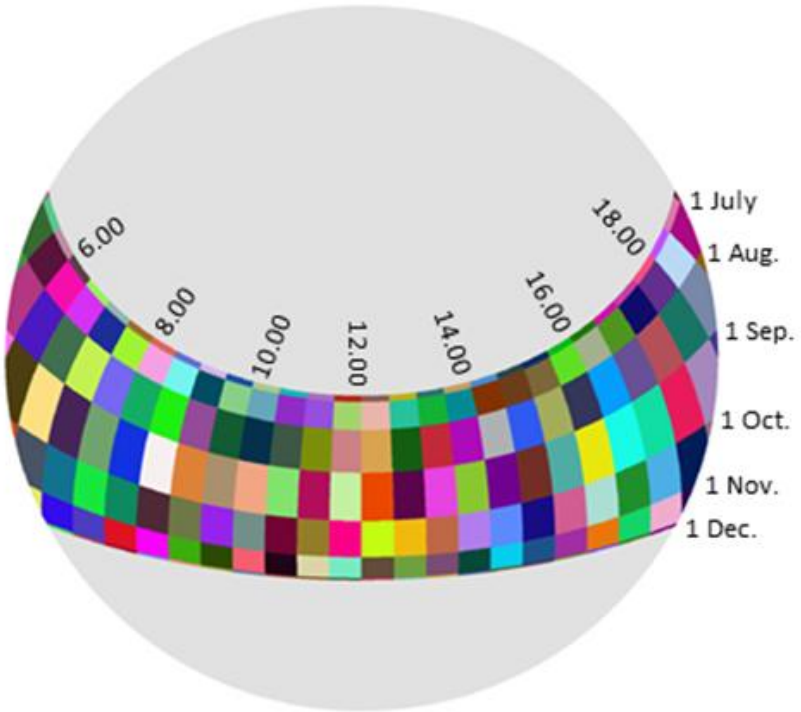


Horizon angles along a profile

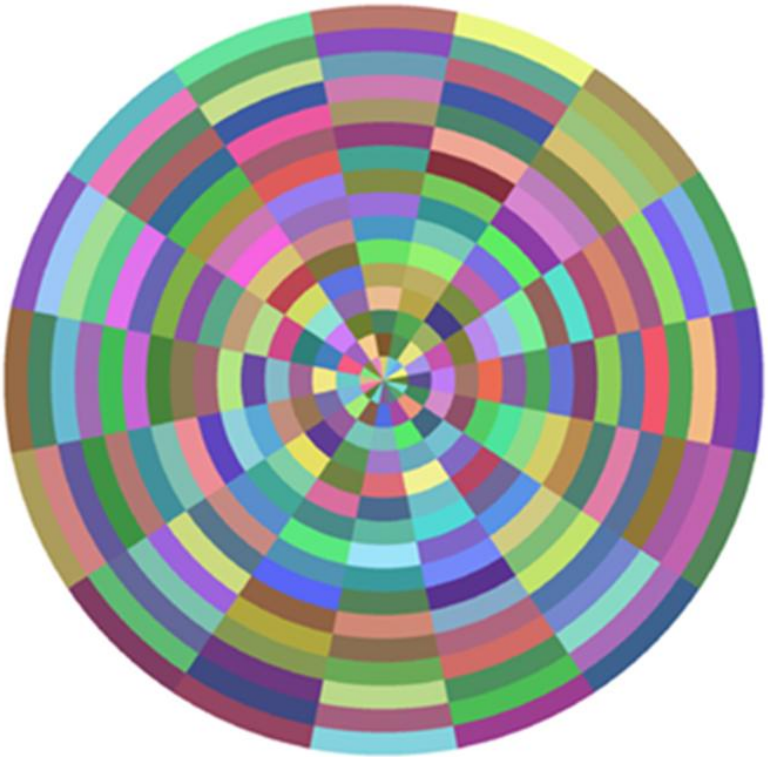


Hemispherical viewshed of a given location (light grey – visible; dark grey – obstructed)

Sunmap and skymap

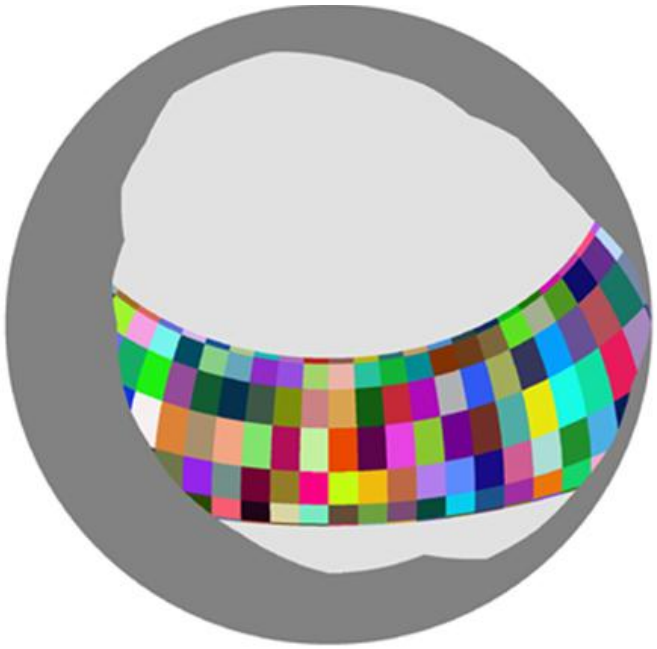


Sumap used for direct radiation (39-deg N)

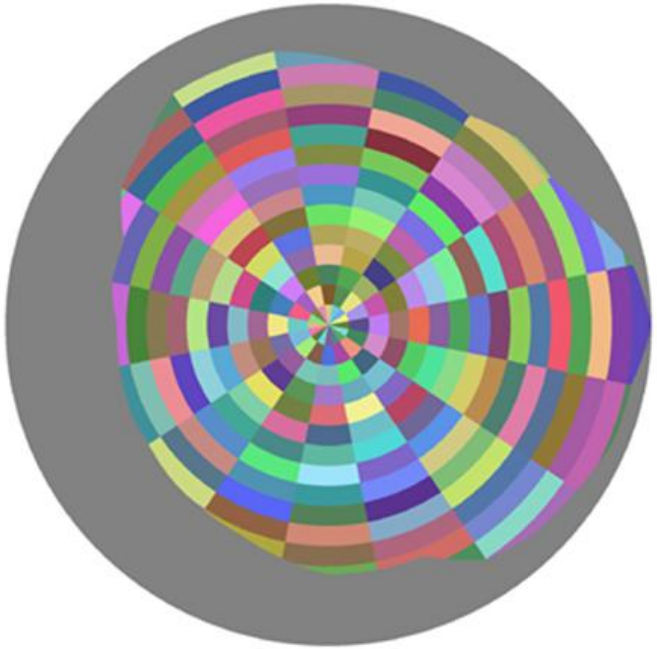


Skymap used for diffuse radiation

Sunmap and skymap in a hemispherical viewshed



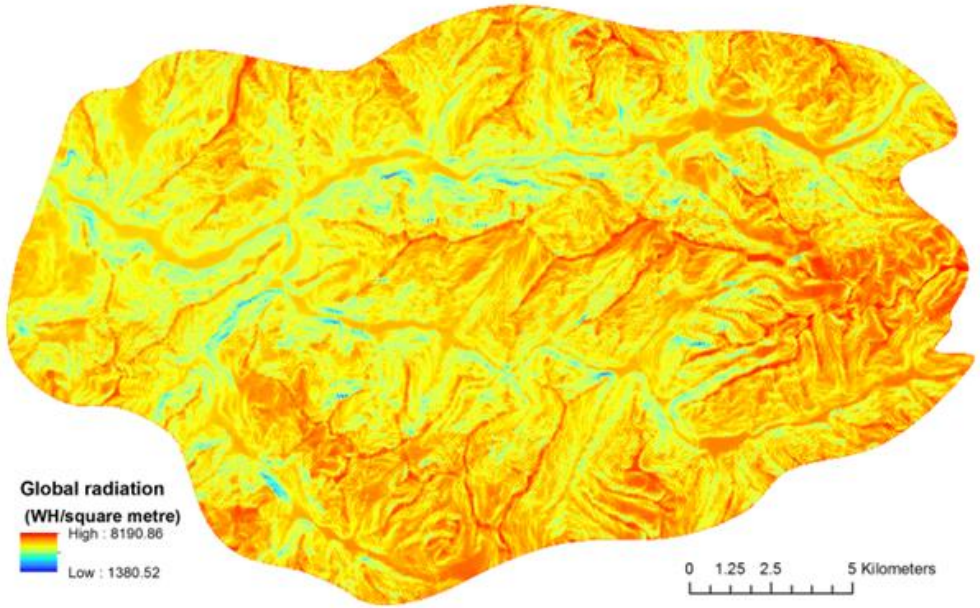
(a)



(b)

Global solar radiation map

Surface solar radiation maps



Rooftop solar power estimates



Homework and Resources

- Read Chapter 8 of course textbook: Terrain Analysis.
- Take the quiz to test your basic understanding (on Canvas).

Further resources

ESRI Training: Terrain Analysis Using ArcGIS Pro

https://www.esri.com/training/catalog/57630436851d31e02a43f18f/terrain-analysis-using-arcgis-pro/

Esri Storymap: Terrain Analysis Case study: Rauriser Ache Catchment

https://storymaps.arcgis.com/stories/d21814c2303448ef84a6d50ae571c24b